DropWise: locating a water-network leak from a few pressure sensors — a pre-registered simulation benchmark and cost model

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1. Aim and hypotheses

Old water networks lose a large share of their water: in the 2026 tariff calculation for Chișinău, losses of 25.9% of the water taken in were accepted [4]. Pressure sensors are cheap, but a crew that digs where the biggest pressure drop appears may open many dry holes before it finds the leak. We tested whether a hydraulic model of the network reduces this waste. The hypotheses and decision rules were fixed before any method was run (repository file *research/PREREGISTRATION.md*, committed before the code that tests it). **H1:** model fitting needs at least 50% fewer dry holes than the naive rule (dig near the sensor with the largest pressure drop) over N = 1,000 random leaks. **H2:** a simple fingerprint method finds the leak within three digs no more than 5 percentage points less often than model fitting (upper end of the 95% interval at most 5 points) and is at least 10 times faster. A hypothesis is supported only if the estimate reaches the target and the 95% bootstrap interval excludes zero.

2. Methodology

Simulator. The pipes of the project's leak hunter form a graph of 14 junctions and 19 pipes with 6 loops (13.4 km; 1 map unit = 4 m). Steady flow is solved with Hazen–Williams head loss and the Global Gradient Algorithm [1] (20 unit tests: mass balance at every junction, the head-loss law in every pipe). A leak is an orifice, $q = k\sqrt{h}$. **Scenarios.** One leak per scenario, at a random place along the pipes, losing 5–25 L/s (2–10% of a 0.25 m³/s demand). The methods do not know the real demand (±10% per junction, ±5% overall). Four pressure sensors sit at junctions chosen by a fixed rule, with error $\sigma = 0.10$ m. A dig finds the leak if it lands within 40 map units (160 m). **Methods.** Each ranks 170 candidate places; the same crew digs down the list (at most 10 digs): random order; naive threshold; *fingerprint matching* (cosine similarity between the readings and pre-computed leak signatures); *model fitting* (Levenberg–Marquardt fit of leak size and overall demand with the full hydraulic model at every place; cf. [2, 3]). **Statistics.** Paired bootstrap, 2,000 resamples. The benchmark seed (2027) was used once; debugging used another seed. **Limits.** A simulation of a drawn network with an exact model, one leak, steady flow and no real field data.

3. Quantitative benchmark results

Table 1. Dry holes before the leak is found (N = 1,000; 10 = not found within 10 digs)

Method	Dry holes (95% CI)	Found at 1st dig	Within 3 digs	Never found	Time per leak
Random order	8.48 (8.30–8.66)	2.8%	8.7%	72.8%	14 µs
Naive threshold	6.91 (6.67–7.15)	10.4%	19.7%	53.7%	52 µs
Fingerprint matching	6.26 (6.00–6.49)	12.2%	25.8%	44.2%	30 µs
Model fitting	5.97 (5.73–6.21)	12.6%	29.3%	38.8%	243 ms

H1 is partially supported: model fitting needed 13.6% fewer dry holes than the naive rule (95% interval 10.0% to 17.2%), far below the 50% target although the interval excludes zero. **H2 is partially supported:** the fingerprint method was 8,035 times faster (0.03 ms against 243 ms) but found the leak within three digs 3.5 points less often than model fitting, with an upper bound of 6.6 points, above the 5-point margin. **Robustness** (300 leaks each, one change at a time) shows what limits the result: the saving against the naive rule is 40.8% for fingerprints and 24.4% for model fitting when demand is known exactly, but only 0.7% and 6.6% when it is twice as uncertain. Model fitting saves 21.2% with 0.03 m sensors and 3.8% with 0.30 m sensors (interval −2.0% to 9.4%), 8.4% with three sensors and 18.0% with six, 26.1% for the largest third of the leaks. A leak on the main line next to the plant is the hardest case (naive rule 9.4 dry holes, worse than random order). Model fitting failed to find 38.8% of the leaks within ten digs. With demand known, the simple

fingerprint method beat model fitting (3.80 against 4.86 dry holes); we suspect the fitted demand factor then absorbs sensor noise, but did not test this.

4. Economic and social applicability

Money, in lei and euro. Tariff 14.03 lei/m³ (Apă-Canal Chișinău, households; press report of the 2026 ANRE decision [4]); €1 = 20.14 lei [5]. The average benchmark leak (12 L/s) loses 1,037 m³ a day, which is 14,546 lei at the full tariff and the daily drinking water of about 7,975 people (130 L each). Valuing a saved m³ at half the tariff, a dig at €800 and half a day per dig, one dry hole costs €981 (19,749 lei). Model fitting avoids 0.94 dry holes per leak, saving €922; with 4 large leaks a year and €1,500 a year of running costs, a €12,000 system of four loggers pays back in 66 months (fingerprints: 138 months); 4.2 large leaks a year give a five-year payback. If customer demand were known, for example from smart meters, the payback falls to 16 months (fingerprints) and 31 months (model fitting). **Comparison.** Retail listings put a 0–10 bar transmitter at €165.99–€215 and a logger with modem at US$1,095–1,925; acoustic-logger kits cost US$13,500–22,500; a satellite survey about US$294 per km (about US$3,921 for this network, a one-off survey rather than a continuous alarm). These prices were seen in web searches and several pages could not be opened. **Assumptions.** The cost of a dig, days per dig, leaks per year and the system cost are assumptions without a source and are the weakest part of the estimate; they are sliders on the project's Research page.

5. Conclusions

With realistic uncertainty (demand ±10% per junction, 0.10 m sensors, four sensors) a hydraulic model finds a leak only modestly better than a simple rule, far from the pre-registered target. The limiting factor is not the algorithm but what is unknown: when demand is known, even the cheap fingerprint method saves 41% of the dry holes and pays back in about 1.4 years at the assumed costs. A utility should therefore invest in demand data and low-noise sensors before complex algorithms. Next steps: calibrate the model on a real Moldovan district, add a flow measurement at the plant and several simultaneous leaks, and replace the cost assumptions with figures from the operator.

References

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2. Pérez R., Sanz G., Puig V., Quevedo J., et al. Leak localization in water networks: a model-based methodology using pressure sensors applied to a real network in Barcelona. IEEE Control Syst Mag 2014;34(4):24–36.
3. Pérez R., Puig V., Pascual J., Quevedo J., Landeros E., Peralta A. Methodology for leakage isolation using pressure sensitivity analysis in water distribution networks. Control Eng Pract 2011;19(10):1157–1167.
4. Logos Press. Tarifele la apă și canalizare cresc pentru locuitorii municipiului Chișinău (ANRE decision), 4 August 2026.
5. National Bank of Moldova. Official exchange rate EUR/MDL, 23 September 2026.

Availability. Code, pre-registration, results and logbook are in the project repository (*research/*, *docs/moldsef/*); every number above is produced by *research/benchmark.ts* and *src/utils/leakEconomics.ts* through *research/docs.ts*.